

AKSHAT ASWAL

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EDUCATION

Graphic Era Hill University, Dehradun

B.Tech in Computer Science & Engineering

Expected Jun 2028

St. Thomas Convent School, Pauri

Senior Secondary (XII) — Science Stream

Apr 2024

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, C++, Python, Java, HTML5, CSS3

Frameworks: Next.js, React.js, Flask, Three.js, React Three Fiber, Material UI, Bootstrap 5

Databases: PostgreSQL, MySQL, SQLite, Supabase

Tools & Platforms: Git, GitHub, Vercel, Netlify, PythonAnywhere, Apache Maven, Apache Tomcat

Cloud & DevOps: CI/CD Pipelines, Row Level Security (RLS), PWA, Capacitor (Android)

Concepts: REST APIs, JWT Authentication, Google OAuth 2.0, MVC Architecture, RBAC, Backtracking Algorithms

PROJECTS

Aegis Intelligence — AI Companion Productivity Platform

Jan 2026 – Present

Next.js 16 · TypeScript · Three.js · React Three Fiber · Supabase · Groq (Llama 3) · ElevenLabs TTS · Capacitor · Vercel

- Built a full-stack AI web app with 3 interactive 3D companions rendered in-browser using Three.js and React Three Fiber, featuring physics-based animation and real-time AI responses via Groq (Llama 3)
- Engineered persistent long-term memory with Supabase PostgreSQL, enabling personalised AI conversations that reference user goals across sessions
- Developed a real-time voice pipeline using ElevenLabs TTS with automatic dual-key failover (20,000 chars/month) and browser Speech API fallback for 100% uptime
- Deployed as a cross-platform app via Next.js (Vercel), PWA manifest, and Capacitor for Android APK — single codebase across browser, desktop, and mobile

Cloud-Based File Storage System

Aug 2025 – Nov 2025

React.js · Supabase · PostgreSQL · Material UI · Netlify · CI/CD

- Built a secure cloud storage platform with role-based access control (RBAC) and a responsive Material UI dashboard supporting multiple user roles
- Designed PostgreSQL schema and implemented Row Level Security (RLS) ensuring complete data isolation across users
- Deployed on Netlify with an automated CI/CD pipeline for continuous delivery and zero-downtime updates

SmartSeat — Intelligent Exam Seating Arrangement System

2025

Python Flask · C++ · HTML5 · CSS Grid · Vanilla JavaScript · SQLite · PythonAnywhere

- Engineered a C++ backtracking algorithm called via Python Flask to generate conflict-free, branch-separated exam seating arrangements at scale
- Implemented zero-adjacency logic and a toggleable strict diagonal separation mode, guaranteeing no two students from the same branch share adjacent seats on all 8 sides
- Built a visual Cinema-Style seating map with real-time async search and CSV bulk import; deployed on PythonAnywhere with SQLite persistent storage

CERTIFICATIONS

Cloud Computing · NPTEL · Issued Oct 2025

Jul – Oct 2025

EXTRACURRICULAR ACTIVITIES

NSS Volunteer — Led community service drives; developed public speaking and team leadership skills while coordinating logistics during social service events